STUDENT NAME AND SURNAME :_____

STUDENT SIGNATURE :_____

Marmara University **Faculty of Engineering**	**EE2001 and EE201 Midterm Exam**	**09.12.2020** **PAGE 1/5**

PLEASE FIRST READ HERE
General Instructions:

- This exam has 5 questions, each in a different page.
- Please **SOLVE THE QUESTIONS WITH YOUR HAND TO EMPTY PAPERS**, write your **name**, **surname** and **signature** to the top of each paper, take their photos or scan them and upload to UZEM-UES system, **until Thursday, 10 December 2020, 11.00 a.m. (in the morning).**
- After each question, a Table including the student specific question parameters is provided. Please find the row with your student number in each table and use the parameters in that row when solving the question.
- If you face any problems or would like to ask any questions about the exam, please send e-mail to: alparslan.zehir@marmara.edu.tr

QUESTION 1/5:

Note: In this question, for $\boldsymbol{V_1}$, $\boldsymbol{A_1}$, $\boldsymbol{T_1}$ and $\boldsymbol{T_2}$, please use the numerical values provided for you in the same row with your student number in Table 1 in the next page.

A battery of $\boldsymbol{V_1}$ volts is charged with a constant current of $\boldsymbol{A_1}$ mA.

(5 pts.) 1.a) How much energy (in J) is transferred in $\boldsymbol{T_1}$ minutes?

(5 pts.) 1.b) How much energy (in kWh) is transferred in $\boldsymbol{T_2}$ hours? (Hint: Energy in kWh is calculated by multiplying the power in kW with the number of hours.)

Table 1. Question 1 Parameters

Student No	Student Surname	QUESTION 1 Parameters			
		V_1	A_1	T_1	T_2
100714004	ATLIHAN	46	21.5	55	20
150116063	OKUMUŞ	198	59.5	359	96
150712026	TÜRKMEN	268	77	499	131
150712030	ERMAN	124	41	211	59
150712049	SALTABAŞ	222	65.5	407	108
150713030	AKBAŞ	24	16	11	9
150714020	AYDIN	52	23	67	23
150714021	AKÖZ	28	17	19	11
150714023	BOSTANCIOĞLU	78	29.5	119	36
150714048	AYDIN	54	23.5	71	24
150715037	PEKTAŞ	220	65	403	107
150715044	ASLAN	38	19.5	39	16
150715046	UYSAL	276	79	515	135
150715050	YILDIZ	290	82.5	543	142
150715052	TANER	242	70.5	447	118
150715053	SERTLEROĞULLARI	226	66.5	415	110
150716001	DAĞDAŞ	102	35.5	167	48
150716009	YÜKCÜ	298	84.5	559	146
150716016	ERYILMAZ	130	42.5	223	62
150716024	YILMAZ	292	83	547	143
150716026	KAHRAMAN	156	49	275	75
150716028	KURMUŞ	180	55	323	87
150716030	AKDEMİR	26	16.5	15	10
150716033	BİLGİÇ	74	28.5	111	34
150716036	CÜCE	96	34	155	45
150716039	ÇOKLUK	98	34.5	159	46
150716043	HALIGÜR	152	48	267	73
150716046	KARACA	162	50.5	287	78
150716057	ULU	272	78	507	133
150716075	KARTAL	170	52.5	303	82
150716501	KAHYA	158	49.5	279	76
150716502	ÖZTÜRK	214	63.5	391	104
150717001	ATAN	42	20.5	47	18
150717004	DAĞLI	104	36	171	49
150717005	CANSIZER	94	33.5	151	44
150717008	BUĞUR	82	30.5	127	38
150717009	EKİCİ	116	39	195	55
150717010	ÖZÇELİK	206	61.5	375	100
150717014	MOR	194	58.5	351	94
150717015	AKTAŞ	30	17.5	23	12
150717019	BİNGÖL	76	29	115	35
150717021	AVCIOĞLU	48	22	59	21
150717022	ÖZCAN	204	61	371	99
150717024	GÖNÜL	138	44.5	239	66

Student No	Student Surname	QUESTION 1 Parameters			
		V_1	A_1	T_1	T_2
150717025	CANBAZ	92	33	147	43
150717031	ÖZTÜRK	216	64	395	105
150717032	BALCI	66	26.5	95	30
150717034	ÖZKUL	210	62.5	383	102
150717036	VURUCU	286	81.5	535	140
150717037	YILMAZ	294	83.5	551	144
150717038	DUMLU	114	38.5	191	54
150717043	TÜRK	264	76	491	129
150717047	ŞAHİN	234	68.5	431	114
150717049	ÖZKAN	208	62	379	101
150717051	PATOĞLU	218	64.5	399	106
150717054	ARSLAN	36	19	35	15
150717055	TEMİZ	248	72	459	121
150717058	ÇÖMEZ	100	35	163	47
150717060	TEKELİ	244	71	451	119
150717853	KOCABAŞ	176	54	315	85
150718003	BOYRAZ	80	30	123	37
150718005	ELBİR	118	39.5	199	56
150718006	SEZGİN	228	67	419	111
150718008	KARAGÖZLÜ	166	51.5	295	80
150718010	GÜNDOĞAN	142	45.5	247	68
150718011	YURDAKUL	296	84	555	145
150718014	DİŞİBÜYÜK	110	37.5	183	52
150718015	BÜYÜKKAĞNICI	84	31	131	39
150718016	TÜRK	266	76.5	495	130
150718017	ŞAHİN	236	69	435	115
150718018	DELİASLAN	106	36.5	175	50
150718019	ABDİ	20	15	3	7
150718020	ŞAHİN	238	69.5	439	116
150718021	GÜNDÜZ	144	46	251	69
150718022	BEDİR	72	28	107	33
150718023	YAVAŞ	288	82	539	141
150718024	KARASLAN	168	52	299	81
150718026	MERT	192	58	347	93
150718028	UZUNER	280	80	523	137
150718029	AYDOĞAN	60	25	83	27
150718031	EROĞLU	126	41.5	215	60
150718033	TEMİR	246	71.5	455	120
150718034	ER	120	40	203	57
150718035	AKYÜREK	32	18	27	13
150718036	SARICAN	224	66	411	109
150718037	ATABAY	40	20	43	17
150718039	KÜÇÜKKÖŞKER	182	55.5	327	88
150718042	AĞCABAY	22	15.5	7	8
150718043	TİRYAKİ	254	73.5	471	124
150718045	KALKIR	160	50	283	77

Student No	Student Surname	QUESTION 1 Parameters			
		V_1	A_1	T_1	T_2
150718046	TUNÇBİLEK	258	74.5	479	126
150718047	MANZAK	188	57	339	91
150718048	KAYA	172	53	307	83
150718049	GÜÇ	140	45	243	67
150718050	KARACA	164	51	291	79
150718051	EROĞLU	128	42	219	61
150718052	TOK	256	74	475	125
150718053	BAYRAM	68	27	99	31
150718055	BEBEK	70	27.5	103	32
150718056	CAFEROĞLU	90	32.5	143	42
150718058	ÖZTOPRAK	212	63	387	103
150718059	SÜNE	232	68	427	113
150718060	DOĞAN	112	38	187	53
150718061	TERZİ	252	73	467	123
150718062	ATASEVER	44	21	51	19
150718501	KİRAZ	174	53.5	311	84
150718503	AYDI	50	22.5	63	22
150718852	ULU	274	78.5	511	134
150719009	NECİP	196	59	355	95
150719011	DENİZ	108	37	179	51
150719015	AYDİN	58	24.5	79	26
150719016	İBİŞ	154	48.5	271	74
150719040	BÜYÜKKAYA	86	31.5	135	40
150719054	GÜRBÜZ	146	46.5	255	70
150719058	GEDİK	136	44	235	65
150719518	ÜNLÜ	282	80.5	527	138
150719523	UZUN	278	79.5	519	136
150719533	AYDIN	56	24	75	25
150719582	MAROUF	190	57.5	343	92
150719583	ALHAJ MASOUD	34	18.5	31	14
150719608	OKURER	202	60.5	367	98
150719621	SÖKMEN	230	67.5	423	112
150719634	ŞAHİN	240	70	443	117
150719662	LADİKLİOĞLU	184	56	331	89
150719663	TURAKOĞLU	260	75	483	127
150719827	FİDAN	134	43.5	231	64
150719828	UĞURLUER	270	77.5	503	132
150719829	ERDEM	122	40.5	207	58
150719841	MAMATİ	186	56.5	335	90
150720821	KOÇMARLI	178	54.5	319	86
150720822	AYGÜN	62	25.5	87	28
150720823	TÜĞMEN	262	75.5	487	128
150818005	HACIOĞLU	148	47	259	71
150818015	BAHÇELİ	64	26	91	29
199520029	VISINI	284	81	531	139
199819097	ESERÇELİK	132	43	227	63

STUDENT NAME AND SURNAME :_____

STUDENT SIGNATURE :_____

QUESTION 2/5:

Note: When solving 2.b part of this question, for V_a, V_b, V_c, V_d, V_e and I_a, I_b, I_c, I_d please use the numerical values provided for you in the same row with your student number in Table 2 in the next page.

The circuit below identifies voltage polarities and current directions for each component.

(5 pts.) 2.a) Using only the voltage polarities and current directions, without any numerical computation, please verbally predict which components absorb power and which components supply power using the passive sign convention.

(5 pts.) 2.b) Using the numerical values given for the voltage and current of each component in Table 2 in the next page, how much total power is absorbed in the circuit?

(5 pts.) 2.c) If the total supplied and absorbed power in this circuit is in balance, what should be the value of the current for component e (I_e)?

(5 pts.) 2.d) Based on the computations in the solution of 2.b and 2.c, identify the components that supply power and those that absorb power. For which components, you found different results than your verbal predictions in your answer for 2.a?

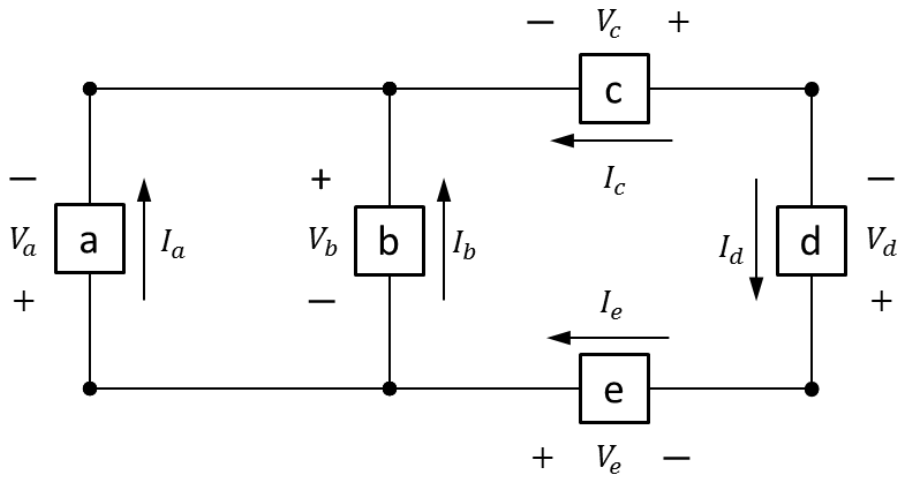


Table 2. Question 2 Parameters

Student No	Student Surname	QUESTION 2 Parameters								
		V_a	V_b	V_c	V_d	V_e	I_a	I_b	I_c	I_d
100714004	ATLIHAN	36	85	102	196	134	-54	30	101	-47
150116063	OKUMUŞ	188	465	406	1108	742	-358	182	633	-275
150712026	TÜRKMEN	258	640	546	1528	1022	-498	252	878	-380
150712030	ERMAN	114	280	258	664	446	-210	108	374	-164
150712049	SALTABAŞ	212	525	454	1252	838	-406	206	717	-311
150713030	AKBAŞ	14	30	58	64	46	-10	8	24	-14
150714020	AYDIN	42	100	114	232	158	-66	36	122	-56
150714021	AKÖZ	18	40	66	88	62	-18	12	38	-20
150714023	BOSTANCIOĞLU	68	165	166	388	262	-118	62	213	-95
150714048	AYDIN	44	105	118	244	166	-70	38	129	-59
150715037	PEKTAŞ	210	520	450	1240	830	-402	204	710	-308
150715044	ASLAN	28	65	86	148	102	-38	22	73	-35
150715046	UYSAL	266	660	562	1576	1054	-514	260	906	-392
150715050	YILDIZ	280	695	590	1660	1110	-542	274	955	-413
150715052	TANER	232	575	494	1372	918	-446	226	787	-341
150715053	SERTLEROĞULLARI	216	535	462	1276	854	-414	210	731	-317
150716001	DAĞDAŞ	92	225	214	532	358	-166	86	297	-131
150716009	YÜKCÜ	288	715	606	1708	1142	-558	282	983	-425
150716016	ERYILMAZ	120	295	270	700	470	-222	114	395	-173
150716024	YILMAZ	282	700	594	1672	1118	-546	276	962	-416
150716026	KAHRAMAN	146	360	322	856	574	-274	140	486	-212
150716028	KURMUŞ	170	420	370	1000	670	-322	164	570	-248
150716030	AKDEMİR	16	35	62	76	54	-14	10	31	-17
150716033	BİLGİÇ	64	155	158	364	246	-110	58	199	-89
150716036	CÜCE	86	210	202	496	334	-154	80	276	-122
150716039	ÇOKLUK	88	215	206	508	342	-158	82	283	-125
150716043	HALIGÜR	142	350	314	832	558	-266	136	472	-206
150716046	KARACA	152	375	334	892	598	-286	146	507	-221
150716057	ULU	262	650	554	1552	1038	-506	256	892	-386
150716075	KARTAL	160	395	350	940	630	-302	154	535	-233
150716501	KAHYA	148	365	326	868	582	-278	142	493	-215
150716502	ÖZTÜRK	204	505	438	1204	806	-390	198	689	-299
150717001	ATAN	32	75	94	172	118	-46	26	87	-41
150717004	DAĞLI	94	230	218	544	366	-170	88	304	-134
150717005	CANSIZER	84	205	198	484	326	-150	78	269	-119
150717008	BUĞUR	72	175	174	412	278	-126	66	227	-101
150717009	EKİCİ	106	260	242	616	414	-194	100	346	-152
150717010	ÖZÇELİK	196	485	422	1156	774	-374	190	661	-287
150717014	MOR	184	455	398	1084	726	-350	178	619	-269
150717015	AKTAŞ	20	45	70	100	70	-22	14	45	-23
150717019	BİNGÖL	66	160	162	376	254	-114	60	206	-92
150717021	AVCIOĞLU	38	90	106	208	142	-58	32	108	-50
150717022	ÖZCAN	194	480	418	1144	766	-370	188	654	-284
150717024	GÖNÜL	128	315	286	748	502	-238	122	423	-185

Student No	Student Surname	QUESTION 2 Parameters								
		V_a	V_b	V_c	V_d	V_e	I_a	I_b	I_c	I_d
150717025	CANBAZ	82	200	194	472	318	-146	76	262	-116
150717031	ÖZTÜRK	206	510	442	1216	814	-394	200	696	-302
150717032	BALCI	56	135	142	316	214	-94	50	171	-77
150717034	ÖZKUL	200	495	430	1180	790	-382	194	675	-293
150717036	VURUCU	276	685	582	1636	1094	-534	270	941	-407
150717037	YILMAZ	284	705	598	1684	1126	-550	278	969	-419
150717038	DUMLU	104	255	238	604	406	-190	98	339	-149
150717043	TÜRK	254	630	538	1504	1006	-490	248	864	-374
150717047	ŞAHİN	224	555	478	1324	886	-430	218	759	-329
150717049	ÖZKAN	198	490	426	1168	782	-378	192	668	-290
150717051	PATOĞLU	208	515	446	1228	822	-398	202	703	-305
150717054	ARSLAN	26	60	82	136	94	-34	20	66	-32
150717055	TEMİZ	238	590	506	1408	942	-458	232	808	-350
150717058	ÇÖMEZ	90	220	210	520	350	-162	84	290	-128
150717060	TEKELİ	234	580	498	1384	926	-450	228	794	-344
150717853	KOCABAŞ	166	410	362	976	654	-314	160	556	-242
150718003	BOYRAZ	70	170	170	400	270	-122	64	220	-98
150718005	ELBİR	108	265	246	628	422	-198	102	353	-155
150718006	SEZGİN	218	540	466	1288	862	-418	212	738	-320
150718008	KARAGÖZLÜ	156	385	342	916	614	-294	150	521	-227
150718010	GÜNDOĞAN	132	325	294	772	518	-246	126	437	-191
150718011	YURDAKUL	286	710	602	1696	1134	-554	280	976	-422
150718014	DİŞİBÜYÜK	100	245	230	580	390	-182	94	325	-143
150718015	BÜYÜKKAĞNICI	74	180	178	424	286	-130	68	234	-104
150718016	TÜRK	256	635	542	1516	1014	-494	250	871	-377
150718017	ŞAHİN	226	560	482	1336	894	-434	220	766	-332
150718018	DELİASLAN	96	235	222	556	374	-174	90	311	-137
150718019	ABDİ	10	20	50	40	30	-2	4	10	-8
150718020	ŞAHİN	228	565	486	1348	902	-438	222	773	-335
150718021	GÜNDÜZ	134	330	298	784	526	-250	128	444	-194
150718022	BEDİR	62	150	154	352	238	-106	56	192	-86
150718023	YAVAŞ	278	690	586	1648	1102	-538	272	948	-410
150718024	KARASLAN	158	390	346	928	622	-298	152	528	-230
150718026	MERT	182	450	394	1072	718	-346	176	612	-266
150718028	UZUNER	270	670	570	1600	1070	-522	264	920	-398
150718029	AYDOĞAN	50	120	130	280	190	-82	44	150	-68
150718031	EROĞLU	116	285	262	676	454	-214	110	381	-167
150718033	TEMİR	236	585	502	1396	934	-454	230	801	-347
150718034	ER	110	270	250	640	430	-202	104	360	-158
150718035	AKYÜREK	22	50	74	112	78	-26	16	52	-26
150718036	SARICAN	214	530	458	1264	846	-410	208	724	-314
150718037	ATABAY	30	70	90	160	110	-42	24	80	-38
150718039	KÜÇÜKKÖŞKER	172	425	374	1012	678	-326	166	577	-251
150718042	AĞCABAY	12	25	54	52	38	-6	6	17	-11
150718043	TİRYAKİ	244	605	518	1444	966	-470	238	829	-359
150718045	KALKIR	150	370	330	880	590	-282	144	500	-218

Student No	Student Surname	QUESTION 2 Parameters								
		V_a	V_b	V_c	V_d	V_e	I_a	I_b	I_c	I_d
150718046	TUNÇBİLEK	248	615	526	1468	982	-478	242	843	-365
150718047	MANZAK	178	440	386	1048	702	-338	172	598	-260
150718048	KAYA	162	400	354	952	638	-306	156	542	-236
150718049	GÜÇ	130	320	290	760	510	-242	124	430	-188
150718050	KARACA	154	380	338	904	606	-290	148	514	-224
150718051	EROĞLU	118	290	266	688	462	-218	112	388	-170
150718052	TOK	246	610	522	1456	974	-474	240	836	-362
150718053	BAYRAM	58	140	146	328	222	-98	52	178	-80
150718055	BEBEK	60	145	150	340	230	-102	54	185	-83
150718056	CAFEROĞLU	80	195	190	460	310	-142	74	255	-113
150718058	ÖZTOPRAK	202	500	434	1192	798	-386	196	682	-296
150718059	SÜNE	222	550	474	1312	878	-426	216	752	-326
150718060	DOĞAN	102	250	234	592	398	-186	96	332	-146
150718061	TERZİ	242	600	514	1432	958	-466	236	822	-356
150718062	ATASEVER	34	80	98	184	126	-50	28	94	-44
150718501	KİRAZ	164	405	358	964	646	-310	158	549	-239
150718503	AYDI	40	95	110	220	150	-62	34	115	-53
150718852	ULU	264	655	558	1564	1046	-510	258	899	-389
150719009	NECİP	186	460	402	1096	734	-354	180	626	-272
150719011	DENİZ	98	240	226	568	382	-178	92	318	-140
150719015	AYDİN	48	115	126	268	182	-78	42	143	-65
150719016	İBİŞ	144	355	318	844	566	-270	138	479	-209
150719040	BÜYÜKKAYA	76	185	182	436	294	-134	70	241	-107
150719054	GÜRBÜZ	136	335	302	796	534	-254	130	451	-197
150719058	GEDİK	126	310	282	736	494	-234	120	416	-182
150719518	ÜNLÜ	272	675	574	1612	1078	-526	266	927	-401
150719523	UZUN	268	665	566	1588	1062	-518	262	913	-395
150719533	AYDIN	46	110	122	256	174	-74	40	136	-62
150719582	MAROUF	180	445	390	1060	710	-342	174	605	-263
150719583	ALHAJ MASOUD	24	55	78	124	86	-30	18	59	-29
150719608	OKURER	192	475	414	1132	758	-366	186	647	-281
150719621	SÖKMEN	220	545	470	1300	870	-422	214	745	-323
150719634	ŞAHİN	230	570	490	1360	910	-442	224	780	-338
150719662	LADİKLİOĞLU	174	430	378	1024	686	-330	168	584	-254
150719663	TURAKOĞLU	250	620	530	1480	990	-482	244	850	-368
150719827	FİDAN	124	305	278	724	486	-230	118	409	-179
150719828	UĞURLUER	260	645	550	1540	1030	-502	254	885	-383
150719829	ERDEM	112	275	254	652	438	-206	106	367	-161
150719841	MAMATİ	176	435	382	1036	694	-334	170	591	-257
150720821	KOÇMARLI	168	415	366	988	662	-318	162	563	-245
150720822	AYGÜN	52	125	134	292	198	-86	46	157	-71
150720823	TÜĞMEN	252	625	534	1492	998	-486	246	857	-371
150818005	HACIOĞLU	138	340	306	808	542	-258	132	458	-200
150818015	BAHÇELİ	54	130	138	304	206	-90	48	164	-74
199520029	VISINI	274	680	578	1624	1086	-530	268	934	-404
199819097	ESERÇELİK	122	300	274	712	478	-226	116	402	-176

STUDENT NAME AND SURNAME :_____

STUDENT SIGNATURE :_____

QUESTION 3/5:

Note: In this question, for $\boldsymbol{p_1}$, $\boldsymbol{p_2}$, $\boldsymbol{p_3}$ and $\boldsymbol{p_4}$, please use the numerical values provided for you in the same row with your student number in Table 3 in the next page.

The terminal voltage ($\boldsymbol{v_t}$) and power ($\boldsymbol{p_t}$) were measured on the device shown below and the recorded values are provided in the table below.

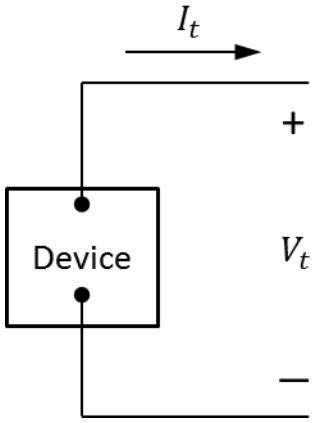
(3 pts.) 3.a) Calculate the terminal current for each measurement.

(3 pts.) 3.b) Using the measured terminal voltages and the calculated terminal currents, draw the line plot of $\boldsymbol{v_t}$ versus $\boldsymbol{i_t}$.

(3 pts.) 3.c) Compute the equation of the line you drew in your answer for 3.b.

(3 pts.) 3.d) Use the equation you computed in your answer for 3.c to construct a circuit model for the device using an ideal voltage source and a resistor.

(3 pts.) 3.e) Use the model you constructed in 3.d to calculate the power the device will deliver to a 15 Ω resistor.



The Table for the power and voltage measurements

$\boldsymbol{v_t}$ (A)	$\boldsymbol{p_t}$ (W)
48	p_1
42	p_2
24	p_3
12	p_4

Table 3. Question 3 Parameters

Student No	Student Surname	QUESTION 3 Parameters			
		p_1	p_2	p_3	p_4
100714004	ATLIHAN	0	117.6	268.8	201.6
150116063	OKUMUŞ	0	756	1728	1296
150712026	TÜRKMEN	0	1050	2400	1800
150712030	ERMAN	0	445.2	1017.6	763.2
150712049	SALTABAŞ	0	856.8	1958.4	1468.8
150713030	AKBAŞ	0	25.2	57.6	43.2
150714020	AYDIN	0	142.8	326.4	244.8
150714021	AKÖZ	0	42	96	72
150714023	BOSTANCIOĞLU	0	252	576	432
150714048	AYDIN	0	151.2	345.6	259.2
150715037	PEKTAŞ	0	848.4	1939.2	1454.4
150715044	ASLAN	0	84	192	144
150715046	UYSAL	0	1083.6	2476.8	1857.6
150715050	YILDIZ	0	1142.4	2611.2	1958.4
150715052	TANER	0	940.8	2150.4	1612.8
150715053	SERTLEROĞULLARI	0	873.6	1996.8	1497.6
150716001	DAĞDAŞ	0	352.8	806.4	604.8
150716009	YÜKCÜ	0	1176	2688	2016
150716016	ERYILMAZ	0	470.4	1075.2	806.4
150716024	YILMAZ	0	1150.8	2630.4	1972.8
150716026	KAHRAMAN	0	579.6	1324.8	993.6
150716028	KURMUŞ	0	680.4	1555.2	1166.4
150716030	AKDEMİR	0	33.6	76.8	57.6
150716033	BİLGİÇ	0	235.2	537.6	403.2
150716036	CÜCE	0	327.6	748.8	561.6
150716039	ÇOKLUK	0	336	768	576
150716043	HALIGÜR	0	562.8	1286.4	964.8
150716046	KARACA	0	604.8	1382.4	1036.8
150716057	ULU	0	1066.8	2438.4	1828.8
150716075	KARTAL	0	638.4	1459.2	1094.4
150716501	KAHYA	0	588	1344	1008
150716502	ÖZTÜRK	0	823.2	1881.6	1411.2
150717001	ATAN	0	100.8	230.4	172.8
150717004	DAĞLI	0	361.2	825.6	619.2
150717005	CANSIZER	0	319.2	729.6	547.2
150717008	BUĞUR	0	268.8	614.4	460.8
150717009	EKİCİ	0	411.6	940.8	705.6
150717010	ÖZÇELİK	0	789.6	1804.8	1353.6
150717014	MOR	0	739.2	1689.6	1267.2
150717015	AKTAŞ	0	50.4	115.2	86.4
150717019	BİNGÖL	0	243.6	556.8	417.6
150717021	AVCIOĞLU	0	126	288	216
150717022	ÖZCAN	0	781.2	1785.6	1339.2
150717024	GÖNÜL	0	504	1152	864

Student No	Student Surname	QUESTION 3 Parameters			
		p_1	p_2	p_3	p_4
150717025	CANBAZ	0	310.8	710.4	532.8
150717031	ÖZTÜRK	0	831.6	1900.8	1425.6
150717032	BALCI	0	201.6	460.8	345.6
150717034	ÖZKUL	0	806.4	1843.2	1382.4
150717036	VURUCU	0	1125.6	2572.8	1929.6
150717037	YILMAZ	0	1159.2	2649.6	1987.2
150717038	DUMLU	0	403.2	921.6	691.2
150717043	TÜRK	0	1033.2	2361.6	1771.2
150717047	ŞAHİN	0	907.2	2073.6	1555.2
150717049	ÖZKAN	0	798	1824	1368
150717051	PATOĞLU	0	840	1920	1440
150717054	ARSLAN	0	75.6	172.8	129.6
150717055	TEMİZ	0	966	2208	1656
150717058	ÇÖMEZ	0	344.4	787.2	590.4
150717060	TEKELİ	0	949.2	2169.6	1627.2
150717853	KOCABAŞ	0	663.6	1516.8	1137.6
150718003	BOYRAZ	0	260.4	595.2	446.4
150718005	ELBİR	0	420	960	720
150718006	SEZGİN	0	882	2016	1512
150718008	KARAGÖZLÜ	0	621.6	1420.8	1065.6
150718010	GÜNDOĞAN	0	520.8	1190.4	892.8
150718011	YURDAKUL	0	1167.6	2668.8	2001.6
150718014	DİŞİBÜYÜK	0	386.4	883.2	662.4
150718015	BÜYÜKKAĞNICI	0	277.2	633.6	475.2
150718016	TÜRK	0	1041.6	2380.8	1785.6
150718017	ŞAHİN	0	915.6	2092.8	1569.6
150718018	DELİASLAN	0	369.6	844.8	633.6
150718019	ABDİ	0	8.4	19.2	14.4
150718020	ŞAHİN	0	924	2112	1584
150718021	GÜNDÜZ	0	529.2	1209.6	907.2
150718022	BEDİR	0	226.8	518.4	388.8
150718023	YAVAŞ	0	1134	2592	1944
150718024	KARASLAN	0	630	1440	1080
150718026	MERT	0	730.8	1670.4	1252.8
150718028	UZUNER	0	1100.4	2515.2	1886.4
150718029	AYDOĞAN	0	176.4	403.2	302.4
150718031	EROĞLU	0	453.6	1036.8	777.6
150718033	TEMİR	0	957.6	2188.8	1641.6
150718034	ER	0	428.4	979.2	734.4
150718035	AKYÜREK	0	58.8	134.4	100.8
150718036	SARICAN	0	865.2	1977.6	1483.2
150718037	ATABAY	0	92.4	211.2	158.4
150718039	KÜÇÜKKÖŞKER	0	688.8	1574.4	1180.8
150718042	AĞCABAY	0	16.8	38.4	28.8
150718043	TİRYAKİ	0	991.2	2265.6	1699.2
150718045	KALKIR	0	596.4	1363.2	1022.4

Student No	Student Surname	QUESTION 3 Parameters			
		p_1	p_2	p_3	p_4
150718046	TUNÇBİLEK	0	1008	2304	1728
150718047	MANZAK	0	714	1632	1224
150718048	KAYA	0	646.8	1478.4	1108.8
150718049	GÜÇ	0	512.4	1171.2	878.4
150718050	KARACA	0	613.2	1401.6	1051.2
150718051	EROĞLU	0	462	1056	792
150718052	TOK	0	999.6	2284.8	1713.6
150718053	BAYRAM	0	210	480	360
150718055	BEBEK	0	218.4	499.2	374.4
150718056	CAFEROĞLU	0	302.4	691.2	518.4
150718058	ÖZTOPRAK	0	814.8	1862.4	1396.8
150718059	SÜNE	0	898.8	2054.4	1540.8
150718060	DOĞAN	0	394.8	902.4	676.8
150718061	TERZİ	0	982.8	2246.4	1684.8
150718062	ATASEVER	0	109.2	249.6	187.2
150718501	KİRAZ	0	655.2	1497.6	1123.2
150718503	AYDI	0	134.4	307.2	230.4
150718852	ULU	0	1075.2	2457.6	1843.2
150719009	NECİP	0	747.6	1708.8	1281.6
150719011	DENİZ	0	378	864	648
150719015	AYDİN	0	168	384	288
150719016	İBİŞ	0	571.2	1305.6	979.2
150719040	BÜYÜKKAYA	0	285.6	652.8	489.6
150719054	GÜRBÜZ	0	537.6	1228.8	921.6
150719058	GEDİK	0	495.6	1132.8	849.6
150719518	ÜNLÜ	0	1108.8	2534.4	1900.8
150719523	UZUN	0	1092	2496	1872
150719533	AYDIN	0	159.6	364.8	273.6
150719582	MAROUF	0	722.4	1651.2	1238.4
150719583	ALHAJ MASOUD	0	67.2	153.6	115.2
150719608	OKURER	0	772.8	1766.4	1324.8
150719621	SÖKMEN	0	890.4	2035.2	1526.4
150719634	ŞAHİN	0	932.4	2131.2	1598.4
150719662	LADİKLİOĞLU	0	697.2	1593.6	1195.2
150719663	TURAKOĞLU	0	1016.4	2323.2	1742.4
150719827	FİDAN	0	487.2	1113.6	835.2
150719828	UĞURLUER	0	1058.4	2419.2	1814.4
150719829	ERDEM	0	436.8	998.4	748.8
150719841	MAMATİ	0	705.6	1612.8	1209.6
150720821	KOÇMARLI	0	672	1536	1152
150720822	AYGÜN	0	184.8	422.4	316.8
150720823	TÜĞMEN	0	1024.8	2342.4	1756.8
150818005	HACIOĞLU	0	546	1248	936
150818015	BAHÇELİ	0	193.2	441.6	331.2
199520029	VISINI	0	1117.2	2553.6	1915.2
199819097	ESERÇELİK	0	478.8	1094.4	820.8

STUDENT NAME AND SURNAME :_____

STUDENT SIGNATURE :_____

QUESTION 4/5:

Note: In this question, for $\boldsymbol{V_1, V_2, I_1, R_1, R_2, R_3}$ and $\boldsymbol{R_L}$ please use the numerical values provided for you
in the same row with your student number in Table 4 in the next page.

For the circuit below:

(10 pts.) 4.a) Use a series of source transformations to find the Thevenin equivalent of the circuit.

(5 pts.) 4.b) Using the Thevenin equivalent you found in your answer for 4.a, find the Norton equivalent.

(5 pts.) 4.c) Find the value of the load resistance R_L that should be connected between the nodes a and b, which results in maximum power being transferred to R_L.

(5 pts.) 4.d) How much current will flow through R_L, when it is connected between the nodes a and b?

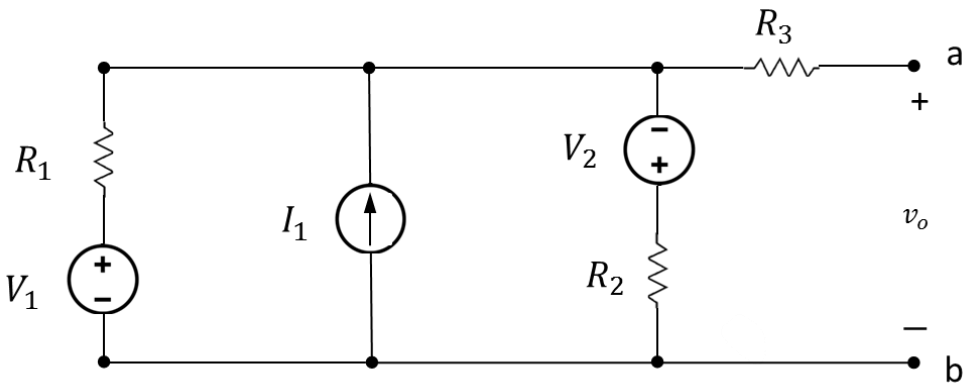


Table 4. Question 4 Parameters

Student No	Student Surname	QUESTION 4 Parameters						
		V_1	V_2	I_1	R_1	R_2	R_3	R_L
100714004	ATLIHAN	46	105	19	41	57	69	80
150116063	OKUMUŞ	198	485	95	269	361	449	536
150712026	TÜRKMEN	268	660	130	374	501	624	746
150712030	ERMAN	124	300	58	158	213	264	314
150712049	SALTABAŞ	222	545	107	305	409	509	608
150713030	AKBAŞ	24	50	8	8	13	14	14
150714020	AYDIN	52	120	22	50	69	84	98
150714021	AKÖZ	28	60	10	14	21	24	26
150714023	BOSTANCIOĞLU	78	185	35	89	121	149	176
150714048	AYDIN	54	125	23	53	73	89	104
150715037	PEKTAŞ	220	540	106	302	405	504	602
150715044	ASLAN	38	85	15	29	41	49	56
150715046	UYSAL	276	680	134	386	517	644	770
150715050	YILDIZ	290	715	141	407	545	679	812
150715052	TANER	242	595	117	335	449	559	668
150715053	SERTLEROĞULLARI	226	555	109	311	417	519	620
150716001	DAĞDAŞ	102	245	47	125	169	209	248
150716009	YÜKCÜ	298	735	145	419	561	699	836
150716016	ERYILMAZ	130	315	61	167	225	279	332
150716024	YILMAZ	292	720	142	410	549	684	818
150716026	KAHRAMAN	156	380	74	206	277	344	410
150716028	KURMUŞ	180	440	86	242	325	404	482
150716030	AKDEMİR	26	55	9	11	17	19	20
150716033	BİLGİÇ	74	175	33	83	113	139	164
150716036	CÜCE	96	230	44	116	157	194	230
150716039	ÇOKLUK	98	235	45	119	161	199	236
150716043	HALIGÜR	152	370	72	200	269	334	398
150716046	KARACA	162	395	77	215	289	359	428
150716057	ULU	272	670	132	380	509	634	758
150716075	KARTAL	170	415	81	227	305	379	452
150716501	KAHYA	158	385	75	209	281	349	416
150716502	ÖZTÜRK	214	525	103	293	393	489	584
150717001	ATAN	42	95	17	35	49	59	68
150717004	DAĞLI	104	250	48	128	173	214	254
150717005	CANSIZER	94	225	43	113	153	189	224
150717008	BUĞUR	82	195	37	95	129	159	188
150717009	EKİCİ	116	280	54	146	197	244	290
150717010	ÖZÇELİK	206	505	99	281	377	469	560
150717014	MOR	194	475	93	263	353	439	524
150717015	AKTAŞ	30	65	11	17	25	29	32
150717019	BİNGÖL	76	180	34	86	117	144	170
150717021	AVCIOĞLU	48	110	20	44	61	74	86
150717022	ÖZCAN	204	500	98	278	373	464	554
150717024	GÖNÜL	138	335	65	179	241	299	356

Student No	Student Surname	QUESTION 4 Parameters						
		V_1	V_2	I_1	R_1	R_2	R_3	R_L
150717025	CANBAZ	92	220	42	110	149	184	218
150717031	ÖZTÜRK	216	530	104	296	397	494	590
150717032	BALCI	66	155	29	71	97	119	140
150717034	ÖZKUL	210	515	101	287	385	479	572
150717036	VURUCU	286	705	139	401	537	669	800
150717037	YILMAZ	294	725	143	413	553	689	824
150717038	DUMLU	114	275	53	143	193	239	284
150717043	TÜRK	264	650	128	368	493	614	734
150717047	ŞAHİN	234	575	113	323	433	539	644
150717049	ÖZKAN	208	510	100	284	381	474	566
150717051	PATOĞLU	218	535	105	299	401	499	596
150717054	ARSLAN	36	80	14	26	37	44	50
150717055	TEMİZ	248	610	120	344	461	574	686
150717058	ÇÖMEZ	100	240	46	122	165	204	242
150717060	TEKELİ	244	600	118	338	453	564	674
150717853	KOCABAŞ	176	430	84	236	317	394	470
150718003	BOYRAZ	80	190	36	92	125	154	182
150718005	ELBİR	118	285	55	149	201	249	296
150718006	SEZGİN	228	560	110	314	421	524	626
150718008	KARAGÖZLÜ	166	405	79	221	297	369	440
150718010	GÜNDOĞAN	142	345	67	185	249	309	368
150718011	YURDAKUL	296	730	144	416	557	694	830
150718014	DİŞİBÜYÜK	110	265	51	137	185	229	272
150718015	BÜYÜKKAĞNICI	84	200	38	98	133	164	194
150718016	TÜRK	266	655	129	371	497	619	740
150718017	ŞAHİN	236	580	114	326	437	544	650
150718018	DELİASLAN	106	255	49	131	177	219	260
150718019	ABDİ	20	40	6	2	5	4	2
150718020	ŞAHİN	238	585	115	329	441	549	656
150718021	GÜNDÜZ	144	350	68	188	253	314	374
150718022	BEDİR	72	170	32	80	109	134	158
150718023	YAVAŞ	288	710	140	404	541	674	806
150718024	KARASLAN	168	410	80	224	301	374	446
150718026	MERT	192	470	92	260	349	434	518
150718028	UZUNER	280	690	136	392	525	654	782
150718029	AYDOĞAN	60	140	26	62	85	104	122
150718031	EROĞLU	126	305	59	161	217	269	320
150718033	TEMİR	246	605	119	341	457	569	680
150718034	ER	120	290	56	152	205	254	302
150718035	AKYÜREK	32	70	12	20	29	34	38
150718036	SARICAN	224	550	108	308	413	514	614
150718037	ATABAY	40	90	16	32	45	54	62
150718039	KÜÇÜKKÖŞKER	182	445	87	245	329	409	488
150718042	AĞCABAY	22	45	7	5	9	9	8
150718043	TİRYAKİ	254	625	123	353	473	589	704
150718045	KALKIR	160	390	76	212	285	354	422

Student No	Student Surname	QUESTION 4 Parameters						
		V_1	V_2	I_1	R_1	R_2	R_3	R_L
150718046	TUNÇBİLEK	258	635	125	359	481	599	716
150718047	MANZAK	188	460	90	254	341	424	506
150718048	KAYA	172	420	82	230	309	384	458
150718049	GÜÇ	140	340	66	182	245	304	362
150718050	KARACA	164	400	78	218	293	364	434
150718051	EROĞLU	128	310	60	164	221	274	326
150718052	TOK	256	630	124	356	477	594	710
150718053	BAYRAM	68	160	30	74	101	124	146
150718055	BEBEK	70	165	31	77	105	129	152
150718056	CAFEROĞLU	90	215	41	107	145	179	212
150718058	ÖZTOPRAK	212	520	102	290	389	484	578
150718059	SÜNE	232	570	112	320	429	534	638
150718060	DOĞAN	112	270	52	140	189	234	278
150718061	TERZİ	252	620	122	350	469	584	698
150718062	ATASEVER	44	100	18	38	53	64	74
150718501	KİRAZ	174	425	83	233	313	389	464
150718503	AYDI	50	115	21	47	65	79	92
150718852	ULU	274	675	133	383	513	639	764
150719009	NECİP	196	480	94	266	357	444	530
150719011	DENİZ	108	260	50	134	181	224	266
150719015	AYDİN	58	135	25	59	81	99	116
150719016	İBİŞ	154	375	73	203	273	339	404
150719040	BÜYÜKKAYA	86	205	39	101	137	169	200
150719054	GÜRBÜZ	146	355	69	191	257	319	380
150719058	GEDİK	136	330	64	176	237	294	350
150719518	ÜNLÜ	282	695	137	395	529	659	788
150719523	UZUN	278	685	135	389	521	649	776
150719533	AYDIN	56	130	24	56	77	94	110
150719582	MAROUF	190	465	91	257	345	429	512
150719583	ALHAJ MASOUD	34	75	13	23	33	39	44
150719608	OKURER	202	495	97	275	369	459	548
150719621	SÖKMEN	230	565	111	317	425	529	632
150719634	ŞAHİN	240	590	116	332	445	554	662
150719662	LADİKLİOĞLU	184	450	88	248	333	414	494
150719663	TURAKOĞLU	260	640	126	362	485	604	722
150719827	FİDAN	134	325	63	173	233	289	344
150719828	UĞURLUER	270	665	131	377	505	629	752
150719829	ERDEM	122	295	57	155	209	259	308
150719841	MAMATİ	186	455	89	251	337	419	500
150720821	KOÇMARLI	178	435	85	239	321	399	476
150720822	AYGÜN	62	145	27	65	89	109	128
150720823	TÜĞMEN	262	645	127	365	489	609	728
150818005	HACIOĞLU	148	360	70	194	261	324	386
150818015	BAHÇELİ	64	150	28	68	93	114	134
199520029	VISINI	284	700	138	398	533	664	794
199819097	ESERÇELİK	132	320	62	170	229	284	338

STUDENT NAME AND SURNAME :_____

STUDENT SIGNATURE :_____

QUESTION 5/5:

Note: In this question, for $\boldsymbol{V_1}$, $\boldsymbol{R_1}$, $\boldsymbol{R_2}$, $\boldsymbol{R_3}$ and $\boldsymbol{R_4}$ please use the numerical values provided for you in the same row with your student number in Table 5 in the next page.

Two circuits are given below:

(15 pts.) 5.a) For the circuit on the left (Figure 5.1) Selecting node a as the reference node, use the node-voltage method to find v_o and the power delivered by the 5 A current source.

(15 pts.) 5.b) For the circuit on the right (Figure 5.2) Use the mesh current method to find the current flowing through $\boldsymbol{R_2}$ resistor**.**

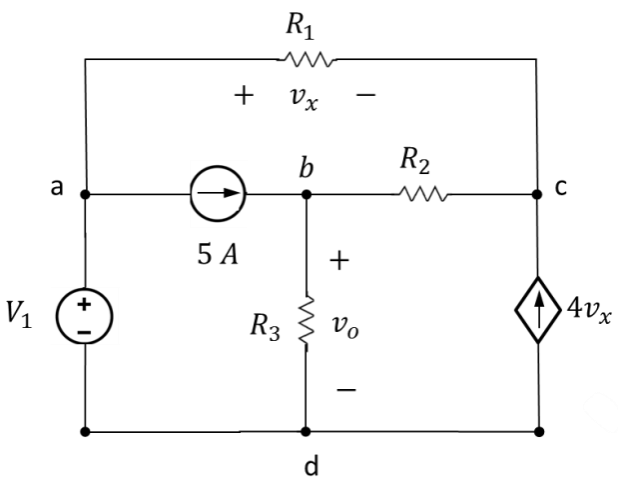


Figure 5.1. Consider this circuit while solving question 5.a

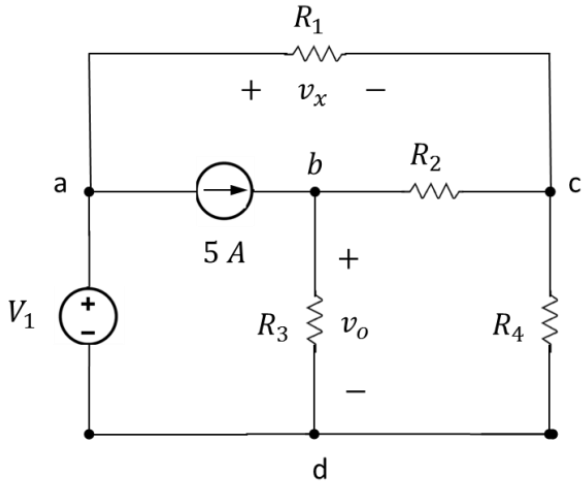


Figure 5.2. Consider this circuit while solving question 5.b

Table 5. Question 5 Parameters

Student No	Student Surname	QUESTION 5 Parameters				
		V_1	R_1	R_2	R_3	R_4
100714004	ATLIHAN	75	15	43	110	55
150116063	OKUMUŞ	455	91	271	718	359
150712026	TÜRKMEN	630	126	376	998	499
150712030	ERMAN	270	54	160	422	211
150712049	SALTABAŞ	515	103	307	814	407
150713030	AKBAŞ	20	4	10	22	11
150714020	AYDIN	90	18	52	134	67
150714021	AKÖZ	30	6	16	38	19
150714023	BOSTANCIOĞLU	155	31	91	238	119
150714048	AYDIN	95	19	55	142	71
150715037	PEKTAŞ	510	102	304	806	403
150715044	ASLAN	55	11	31	78	39
150715046	UYSAL	650	130	388	1030	515
150715050	YILDIZ	685	137	409	1086	543
150715052	TANER	565	113	337	894	447
150715053	SERTLEROĞULLARI	525	105	313	830	415
150716001	DAĞDAŞ	215	43	127	334	167
150716009	YÜKCÜ	705	141	421	1118	559
150716016	ERYILMAZ	285	57	169	446	223
150716024	YILMAZ	690	138	412	1094	547
150716026	KAHRAMAN	350	70	208	550	275
150716028	KURMUŞ	410	82	244	646	323
150716030	AKDEMİR	25	5	13	30	15
150716033	BİLGİÇ	145	29	85	222	111
150716036	CÜCE	200	40	118	310	155
150716039	ÇOKLUK	205	41	121	318	159
150716043	HALIGÜR	340	68	202	534	267
150716046	KARACA	365	73	217	574	287
150716057	ULU	640	128	382	1014	507
150716075	KARTAL	385	77	229	606	303
150716501	KAHYA	355	71	211	558	279
150716502	ÖZTÜRK	495	99	295	782	391
150717001	ATAN	65	13	37	94	47
150717004	DAĞLI	220	44	130	342	171
150717005	CANSIZER	195	39	115	302	151
150717008	BUĞUR	165	33	97	254	127
150717009	EKİCİ	250	50	148	390	195
150717010	ÖZÇELİK	475	95	283	750	375
150717014	MOR	445	89	265	702	351
150717015	AKTAŞ	35	7	19	46	23
150717019	BİNGÖL	150	30	88	230	115
150717021	AVCIOĞLU	80	16	46	118	59
150717022	ÖZCAN	470	94	280	742	371
150717024	GÖNÜL	305	61	181	478	239

Student No	Student Surname	QUESTION 5 Parameters				
		V_1	R_1	R_2	R_3	R_4
150717025	CANBAZ	190	38	112	294	147
150717031	ÖZTÜRK	500	100	298	790	395
150717032	BALCI	125	25	73	190	95
150717034	ÖZKUL	485	97	289	766	383
150717036	VURUCU	675	135	403	1070	535
150717037	YILMAZ	695	139	415	1102	551
150717038	DUMLU	245	49	145	382	191
150717043	TÜRK	620	124	370	982	491
150717047	ŞAHİN	545	109	325	862	431
150717049	ÖZKAN	480	96	286	758	379
150717051	PATOĞLU	505	101	301	798	399
150717054	ARSLAN	50	10	28	70	35
150717055	TEMİZ	580	116	346	918	459
150717058	ÇÖMEZ	210	42	124	326	163
150717060	TEKELİ	570	114	340	902	451
150717853	KOCABAŞ	400	80	238	630	315
150718003	BOYRAZ	160	32	94	246	123
150718005	ELBİR	255	51	151	398	199
150718006	SEZGİN	530	106	316	838	419
150718008	KARAGÖZLÜ	375	75	223	590	295
150718010	GÜNDOĞAN	315	63	187	494	247
150718011	YURDAKUL	700	140	418	1110	555
150718014	DİŞİBÜYÜK	235	47	139	366	183
150718015	BÜYÜKKAĞNICI	170	34	100	262	131
150718016	TÜRK	625	125	373	990	495
150718017	ŞAHİN	550	110	328	870	435
150718018	DELİASLAN	225	45	133	350	175
150718019	ABDİ	10	2	4	6	3
150718020	ŞAHİN	555	111	331	878	439
150718021	GÜNDÜZ	320	64	190	502	251
150718022	BEDİR	140	28	82	214	107
150718023	YAVAŞ	680	136	406	1078	539
150718024	KARASLAN	380	76	226	598	299
150718026	MERT	440	88	262	694	347
150718028	UZUNER	660	132	394	1046	523
150718029	AYDOĞAN	110	22	64	166	83
150718031	EROĞLU	275	55	163	430	215
150718033	TEMİR	575	115	343	910	455
150718034	ER	260	52	154	406	203
150718035	AKYÜREK	40	8	22	54	27
150718036	SARICAN	520	104	310	822	411
150718037	ATABAY	60	12	34	86	43
150718039	KÜÇÜKKÖŞKER	415	83	247	654	327
150718042	AĞCABAY	15	3	7	14	7
150718043	TİRYAKİ	595	119	355	942	471
150718045	KALKIR	360	72	214	566	283

Student No	Student Surname	QUESTION 5 Parameters				
		V_1	R_1	R_2	R_3	R_4
150718046	TUNÇBİLEK	605	121	361	958	479
150718047	MANZAK	430	86	256	678	339
150718048	KAYA	390	78	232	614	307
150718049	GÜÇ	310	62	184	486	243
150718050	KARACA	370	74	220	582	291
150718051	EROĞLU	280	56	166	438	219
150718052	TOK	600	120	358	950	475
150718053	BAYRAM	130	26	76	198	99
150718055	BEBEK	135	27	79	206	103
150718056	CAFEROĞLU	185	37	109	286	143
150718058	ÖZTOPRAK	490	98	292	774	387
150718059	SÜNE	540	108	322	854	427
150718060	DOĞAN	240	48	142	374	187
150718061	TERZİ	590	118	352	934	467
150718062	ATASEVER	70	14	40	102	51
150718501	KİRAZ	395	79	235	622	311
150718503	AYDI	85	17	49	126	63
150718852	ULU	645	129	385	1022	511
150719009	NECİP	450	90	268	710	355
150719011	DENİZ	230	46	136	358	179
150719015	AYDİN	105	21	61	158	79
150719016	İBİŞ	345	69	205	542	271
150719040	BÜYÜKKAYA	175	35	103	270	135
150719054	GÜRBÜZ	325	65	193	510	255
150719058	GEDİK	300	60	178	470	235
150719518	ÜNLÜ	665	133	397	1054	527
150719523	UZUN	655	131	391	1038	519
150719533	AYDIN	100	20	58	150	75
150719582	MAROUF	435	87	259	686	343
150719583	ALHAJ MASOUD	45	9	25	62	31
150719608	OKURER	465	93	277	734	367
150719621	SÖKMEN	535	107	319	846	423
150719634	ŞAHİN	560	112	334	886	443
150719662	LADİKLİOĞLU	420	84	250	662	331
150719663	TURAKOĞLU	610	122	364	966	483
150719827	FİDAN	295	59	175	462	231
150719828	UĞURLUER	635	127	379	1006	503
150719829	ERDEM	265	53	157	414	207
150719841	MAMATİ	425	85	253	670	335
150720821	KOÇMARLI	405	81	241	638	319
150720822	AYGÜN	115	23	67	174	87
150720823	TÜĞMEN	615	123	367	974	487
150818005	HACIOĞLU	330	66	196	518	259
150818015	BAHÇELİ	120	24	70	182	91
199520029	VISINI	670	134	400	1062	531
199819097	ESERÇELİK	290	58	172	454	227